



DevOps



Nexus Repository Manager

Nexus Repository Manager OSS: (Open Source Software)

- Nexus Repository OSS is an open source repository that supports many artifact formats, including Docker, Java™, and npm. With the Nexus tool integration, pipelines in your toolchain can publish and retrieve versioned apps and their dependencies by using central repositories that are accessible from other environments.
- Few of the other features of a repository manager are:
 - 👉 Saving time and bandwidth due to reduced number of downloads off remote repositories.
 - 👉 Backup and Restore
 - 👉 Cleanup Policies
 - 👉 Search Functionality
 - 👉 Multi-format support

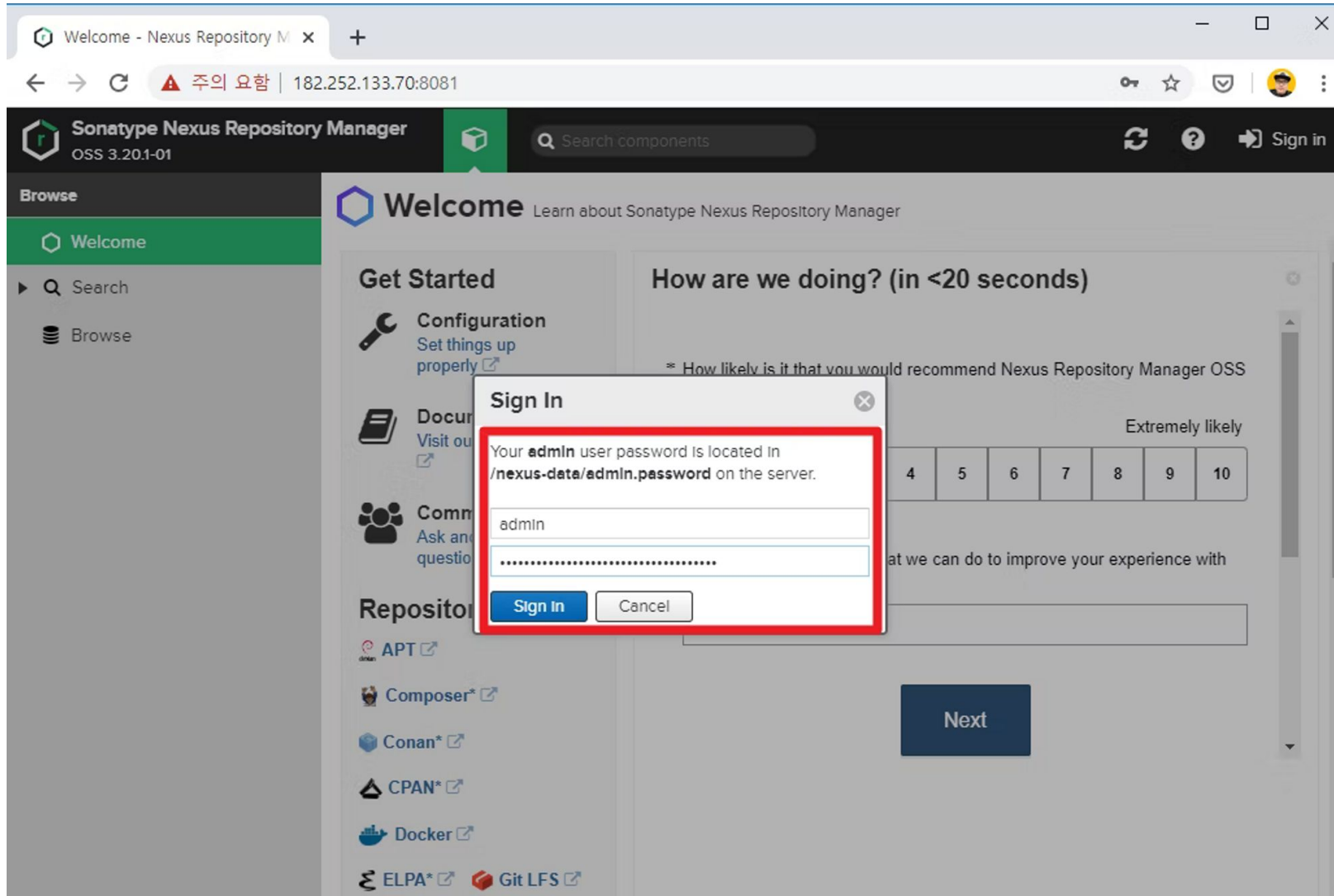
NXRM Installation

```
$ sudo mkdir /nexus-data  
$ sudo chown -R 200:200 /nexus-data # The default group for Nexus is group id 200.  
$ docker run -d -p 8081:8081 -p 5000:5000 --restart always --name nexus -v /nexus-data:/nexus-data  
-u root sonatype/nexus3  
# User: admin & Password is /nexus-data/admin.password
```

NOTE

- Two ports are exposed. On port **8081** the web GUI of Nexus 3 will be available. The Docker registry API will be exposed on port 5000. The latter port may be changed as desired, just remember to configure Nexus 3 accordingly.

NXRM Getting Started



NXRM Getting Started

The image shows the Sonatype Nexus Repository Manager (NXRM) web interface. The top navigation bar includes the Sonatype logo, version 'OSS 3.20.1-01', a search bar, and user controls for 'admin' and 'Sign out'. The left sidebar has a 'Browse' section with links to 'Welcome', 'Search', 'Browse', and 'Upload'. The main content area displays a 'Welcome' message and a 'Get Started' section with links to 'Configuration' and 'Documentation'. A modal dialog is open in the foreground, titled 'Please choose a password for the admin user' (2 of 4). The modal contains two input fields: 'New password:' and 'Confirm password:', both masked with dots. Below the fields are 'Back' and 'Next' buttons. In the background, a survey titled 'How are we doing? (in <20 seconds)' is visible, asking for a recommendation score from 0 to 10.

Sonatype Nexus Repository Manager
OSS 3.20.1-01

Search components

admin Sign out

Welcome Learn about Sonatype Nexus Repository Manager

Get Started

- Configuration**
Set things up properly
- Documentation**
Visit our help site

How are we doing? (in <20 seconds)

⌘ How likely is it that you would recommend Nexus Repository Manager OSS to a friend or colleague?

Not at all likely Extremely likely

0 1 2 3 4 5 6 7 8 9 10

Please choose a password for the admin user 2 of 4

New password:

Confirm password:

Back Next

CPAN* Docker ELPA* Git LFS

NXRM User and Role

- Create Role

Edit audit

Nexus Role

Role Setup

Role ID *

Role Name *

Role Description

Privileges

Available

[+ Transfer All](#)

+ nx-all

+ nx-analytics-all

Given

[* Remove All](#)

* nx-repository-view-*-*browse

* nx-repository-view-*-*read

Available

[+ Transfer All](#)

+ nx-admin

1 item available

Contained

[* Remove All](#)

* nx-anonymous

1 item transferred

NXRM User and Role

- Create User

 **Users** /  **dara dara**

 **Delete user**  **Change password**

Settings

ID:
dara

First name:
dara

Last name:
dara

Email:
Used for notifications
dara@gmail.com

Status:
Active

Roles:

Available

Filter

nx-admin
nx-anonymous

Granted

audit

>
<

Save **Discard**

NXRM Blob Store

Sonatype Nexus Repository Manager
OSS 3.20.1-01

Administration

- Repository
 - Repositories
 - Blob Stores**
 - Content Selectors
 - Cleanup Policies
 - Routing Rules
- Security
 - Privileges
 - Roles
 - Users
 - Anonymous
 - LDAP
 - Realms
 - SSL Certificates
 - IQ Server

Blob Stores / Create blob store

Type: File

Name: docker-hosted

State:
☐ Enable Soft Quota

Path: /nexus-data/blobs/docker-hosted

Create blob store Cancel

NXRM Docker Registry (hosted)

 **Repositories** /  docker-hosted

 Delete repository  Rebuild index

Settings

Name: docker-hosted

Format: docker

Type: hosted

URL: http://182.252.133.70:8081/repository/docker-hosted/

Online: ☒ If checked, the repository accepts incoming requests

Repository Connectors

Connectors allow Docker clients to connect directly to hosted registries, but are not always required. Consult our [documentation](#) for which connector is appropriate for your use case. For information on scaling the repositories see our [scaling documentation](#).

HTTP:

Create an HTTP connector at specified port. Normally used if the server is behind a secure proxy.

☒ 5000

HTTPS:

Create an HTTPS connector at specified port. Normally used if the server is configured for https.

☐

Allow anonymous docker pull:

☐ Allow anonymous docker pull (Docker Bearer Token Realm required)

Docker Registry API Support

Enable Docker V1 API:

☒ Allow clients to use the V1 API to interact with this repository

Storage

Blob store:

docker-hosted

Strict Content Type Validation:

☒ Validate that all content uploaded to this repository is of a MIME type appropriate for the repository format

Hosted

Deployment policy:

Controls if deployments of and updates to artifacts are allowed

Allow redeploy

Cleanup

Cleanup Policies:

Components that match any of the Applied policies will be deleted

Available

Filter

Applied

>

<

Save

Discard

NXRM Docker Registry with Http

```
$ sudo vim /etc/docker/daemon.json  
  
{  
  "insecure-registries" : ["34.124.225.71:5000"]  
}
```

The screenshot displays the Sonatype Nexus Repository Manager (NXRM) administration interface. The top navigation bar includes the Sonatype logo, version 'OSS 3.201-01', a search bar, and a settings icon (gear) highlighted with a red box. The left sidebar shows the 'Administration' menu with categories like Repository, Security, and Realms. The 'Realms' option is highlighted with a red box. The main content area is titled 'Realms' and 'Manage the active security realms and their order'. It features two columns: 'Available' and 'Active'. In the 'Available' column, the 'Docker Bearer Token Realm' is highlighted with a red box. In the 'Active' column, 'Local Authenticating Realm' and 'Local Authorizing Realm' are listed. A red box highlights the right arrow button between the columns, with a yellow 'Add to Selected' tooltip. At the bottom, 'Save' and 'Discard' buttons are visible.

NXRM Docker Registry with Domain Name

```
server {
    listen 80;
    listen [::]:80;

    server_name nexus-ui.e-crops.co;

    location / {
        proxy_pass http://localhost:8081/;
        proxy_set_header Host $http_host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

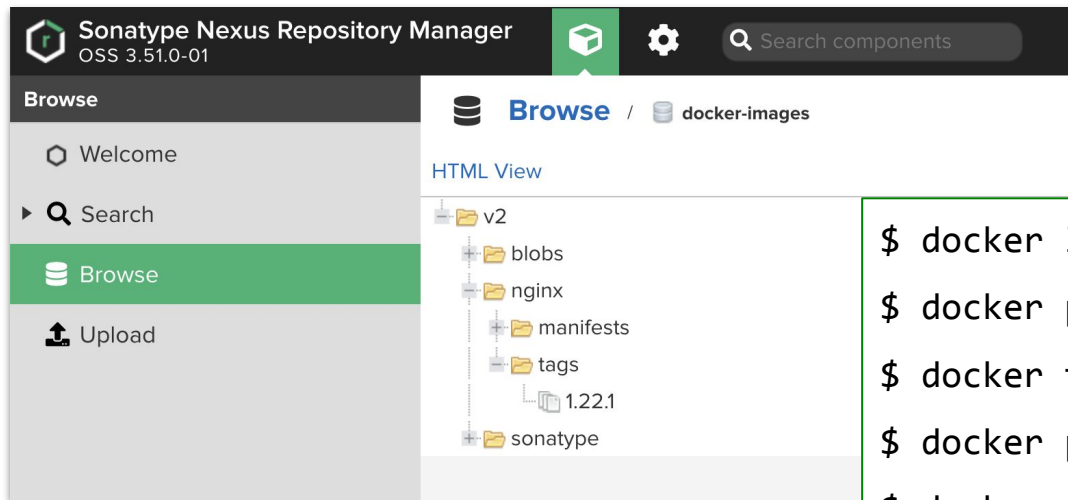
```
server {
    listen 80;
    listen [::]:80;

    server_name nexus.e-crops.co;

    location / {
        proxy_pass http://localhost:5000/;
        proxy_set_header Host $http_host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

NXRM Docker Registry Usage

```
$ docker login -u admin -p 123 34.124.225.71:5000
$ docker pull nginx:1.22.1 # From docker hub
$ docker tag nginx:1.22.1 34.124.225.71:5000/my-nginx:1.22.1
$ docker push 34.124.225.71:5000/my-nginx:1.22.1
$ docker pull 34.124.225.71:5000/my-nginx:1.22.1
```



```
$ docker login -u admin -p 123 nexus.e-crops.co
$ docker pull nginx:1.22.1 # From docker hub
$ docker tag nginx:1.22.1 nexus.e-crops.co/my-nginx:1.22.1
$ docker push nexus.e-crops.co/my-nginx:1.22.1
$ docker pull nexus.e-crops.co/my-nginx:1.22.1
```

NXRM Helm Repository (hosted)

The screenshot shows the Sonatype Nexus Repository Manager (OSS 3.51.0-01) Administration page for the 'helm-hosted' repository. The left sidebar contains the 'Administration' menu with options like Repository, Repositories, Blob Stores, Log4j Visualizer, Proprietary Repositories, Content Selectors, Cleanup Policies, Routing Rules, Security, Privileges, Roles, Users, Anonymous Access, and LDAP. The main content area shows the 'helm-hosted' repository settings, including Name, Format, Type, URL, Online status, Storage (Blob store), Strict Content Type Validation, Hosted (Deployment policy), and Proprietary Components.

Sonatype Nexus Repository Manager
OSS 3.51.0-01

Administration

- Repository
- Repositories**
- Blob Stores
- Log4j Visualizer
- Proprietary Repositories
- Content Selectors
- Cleanup Policies
- Routing Rules
- Security
 - Privileges
 - Roles
 - Users
 - Anonymous Access
 - LDAP

Repositories / **helm-hosted**

Delete repository Rebuild index

Settings

Name: helm-hosted
Format: helm
Type: hosted
URL: http://localhost:8081/repository/helm-hosted/
Online: ☒ If checked, the repository accepts incoming requests

Storage

Blob store: helm-hosted

Strict Content Type Validation:
☒ Validate that all content uploaded to this repository is of a MIME type appropriate for the format

Hosted

Deployment policy:
Controls if deployments of and updates to artifacts are allowed
Allow redeploy

Proprietary Components:
☐ Components in this repository count as proprietary for namespace conflict attacks (requires Sonatype Nexus Firewall)

The screenshot shows the Sonatype Nexus Repository Manager (OSS 3.51.0-01) Browse page for the 'helm-hosted' repository. The left sidebar contains the 'Browse' menu with options like Welcome, Search, Browse, and Upload. The main content area shows the repository structure with a folder 'hello-world' containing a sub-folder '0.1.0' with files 'example.tgz' and 'index.yaml'. There are buttons for 'Upload component' and 'HTML View'.

Sonatype Nexus Repository Manager
OSS 3.51.0-01

Browse

- Welcome
- Search
- Browse**
- Upload

Browse / **helm-hosted**

Upload component HTML View

hello-world

0.1.0

example.tgz

index.yaml

NXRM Helm Repository (hosted)

```
$ git clone https://github.com/helm/examples.git
$ tar -cvzf example.tgz example
$ curl -u admin:123 https://nexus-ui.e-crops.co/repository/helm-hosted/ --upload-file example.tgz
```

Using Helm Client

Once you have Helm up and running you'll want to run a command similar to the following to add a Helm repo:

```
helm repo add <helm_repository_name> http://<host>:<port>/repository/<nexus_repository_name>/ --username <usern
```

The below command will fetch the latest chart or with the version:

1. `helm fetch <helm_repository_name>/<chart_name>`
2. `helm fetch <helm_repository_name>/<chart_name> --version <chart_version>`

A large, faint watermark of the ISTAD logo is centered in the background. It is a circular emblem with a blue outer ring containing the text 'INSTITUTE OF SCIENCE AND TECHNOLOGY ADVANCED DEVELOPMENT' and Khmer text. Inside the ring is a blue circle with a white atomic symbol. The word 'ISTAD' is written in red across the bottom of the inner circle. A blue speech bubble with three white dots is positioned over the center of the logo.

Thank you

Begin with the theory, Start with the practical